Onset of Chaos in dissipative cat qubit

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Chaos in quantum computing is seldom a feature that is looked for: it is much harder to stabilize any state and you loose all memory of previous computation step. This is why it is crucial to investigate its onset for any architecture, and if chaos there is, how could we mitigate it. In this paper we focus on the autoparametric dissipative cat qubit, show how it can be driven into chaos, and how to partially avoid it. We set an upper threshold on the two-photon rate and buffer amplitude (cat size) and display ways to push this threshold away. Along the way, we showcase a pipeline to study chaos; the tools presented here are versatile and can be used for any Floquet systems.

I. INTRODUCTION

In order to build a useful quantum computer, one need a system which is at the same time stable, to be able to store quantum states, and non-linear, so as to perform non trivial operations. Unfortunately non-linear systems have the tendency to develop chaotic behaviors when their parameters are tuned. First investigations on chaos have been carried in chains of transmons [1]. A few years later in single qubit system such as transmons [2] and Kerr cat qubit [3] were demonstrated to host chaotic dynamics. A Kerr cat is made out of a single mode and quartic non-linearity and thus it is logical to expect dissipative cat qubits, which are made out of two-modes and a transcendental non-linearity to be able to develop chaos. [TMD: Cite Burgelman]

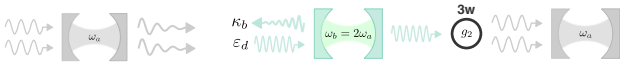
This paper not only show that chaos can develop in a specific dissipative cat qubit, but also showcase a general method to study the onset of chaos in periodic open quantum systems.

In the first part, we present the dissipative cat qubit system that we investigate. In the second part, we carry a classical analysis of this system and in the third part, we focus on the quantum analysis. Finally, we tackle two ways to mitigate chaos in dissipative cat qubit.

II. MODEL / SETUP

Dissipative cat qubits is a transatlantic superconducting architecture which has been developed in the late 2010s at Yale University [4] and ENS Paris [5].

The goal of the dissipative qubit is to stabilize two coherent states $|+\alpha\rangle$ and $|-\alpha\rangle$ in quantum phase space.



a) Diagram of the one-mode cat

b) Diagram of the autoparametric two-mode cat

FIG. 1: Diagrams of different cat architectures

This is achieved through the two-photon loss Lindbladian

$$\mathcal{L} \propto \mathcal{D}[\hat{a}^2 - \alpha^2]. \tag{1}$$

With this Lindbladian, one of the two errors in quantum computation can be suppressed: when the cat-size α is increased, the bit-flip rate decreases exponentially, while the phase-flip rate only increases linearly [5]. Such Lindbladian can be effectively engineered if you add a second mode. The mode a stabilizing the cat state is named "memory" and the second mode b named "buffer": the buffer is leaky and a 2-to-1 photon conversion coupling between the memory and the buffer is implemented Fig. 1. For the photon conversion, a three-wave mixing process, to work we need to account for energy conservation: we opted for the autoparametric architecture [6] where the frequency of the buffer is tailored to match twice the frequency of the memory, but one is free to add some pump to account for any frequency disparities [TMD: Add citations on pumped cat from Alice & Bob and Quantic ENS team]. The reason for this choice is that the autoparametric cat is a Floquet system and is thus easier to analyze.

Such qubit is built with the three musketeers of superconducting circuits: capacitor, inductors and Josephson junctions. In an ideal world, this last element brings all the non-linearity in the system thanks to the relation between voltage V and phase ϕ

$$V \propto \sin(\phi/\Phi_0), \tag{2}$$

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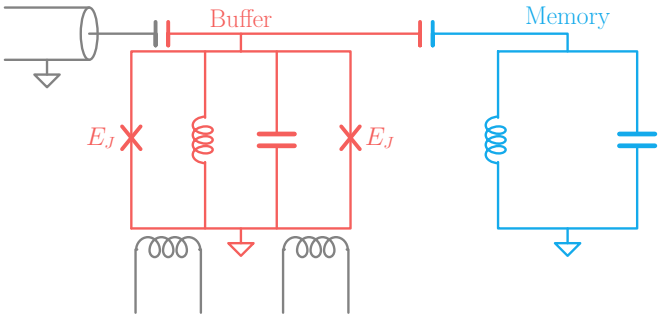


FIG. 2: Example of circuit implementing the autoparametric cat qubit.

where $\Phi_0 = \hbar/2e$ is the reduced magnetic flux quantum. The Hamiltonian of such a circuit contains transcendental (trigonometric) terms and any polynomial expression is only an approximation.

In this paper, we researchers living in an ideal world, we focus on cat qubit without any parasitic terms, but due to some personal repulsion for approximations, we will keep the whole nonlinearity.

The Hamiltonian of a dissipative cat qubit reads [7]

$$\begin{aligned} H = &\omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} + \varepsilon_d \cos(\omega_d t)(\hat{b} + \hat{b}^\dagger) \\ &- g \sin(\hat{\varphi}_a + \hat{\varphi}_b + \varphi_d). \end{aligned} \tag{3}$$

We opted the flux φ_d such that we cancel the linear term of the sine coupling

$$\begin{aligned} H = &\omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} + \varepsilon_d \cos(\omega_d t)(\hat{b} + \hat{b}^\dagger) \\ &- g\big[\sin(\hat{\varphi}_a + \hat{\varphi}_b) - \hat{\varphi}_a - \hat{\varphi}_b\big], \end{aligned} \tag{4}$$

to minimize the frequency shift as the coupling rate g is varied. The dissipation is also considered as ideal, meaning that there is only single-photon loss on the buffer

$$L = \kappa_b \mathcal{D}[b]. \tag{5}$$

Previous works have investigated the impact of loss on the memory on the dynamics[7] or on the bit-flip time[8].

III. CLASSICAL ANALYSIS

In order to build some intuition, we first look at the classical dynamics of Eq. (4). There are two ways to derive the equations that we studied. The first one consist of considering the quantum Hamiltonian as a classical Hamiltonian by asking the operators to kindly doff their hats, then we derive Hamilton equation and we add decay on one of the quadrature. The second way is to start with the GKSL-equation, derive the equation on the Husimi Q-function, approximate it by an isotropic normal gaussian and write down the equation on the mean (Appendix A).

$$\begin{cases} \dot{a} &= -i\omega_a a + ig\,\varphi_a\big(\cos X - 1\big), \\ \dot{b} &= -\Big(i\omega_b + \frac{\kappa_b}{2}\Big)b - i\varepsilon_d \cos(\omega_d t) \\ &\quad + ig\,\varphi_b\big(\cos X - 1\big), \end{cases} \tag{6}$$

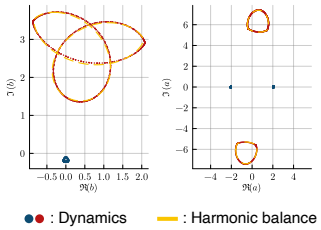


FIG. 3: Limit cycles for the buffer phase space (left) and the memory phase space (right): blue dots corresponds to the cat limit cycles, found by letting trajectories converge to them. The red dots represent the satellite limit cycles, also found by letting trajectories converge in phases space. The yellow dashed line corresponds to the prediction of the harmonic balance.

with $X = \varphi_a(a + a^*) + \varphi_b(b + b^*)$.

A. Phase-space trajectories

The first analysis consists of launching many trajectories in phase-space and study their behavior: do some diverge? do some converge? The trajectories are shown on Fig. 3.Should we consider the RWA cat, all the trajectories ends up either in $|\alpha, 0\rangle$ or $|-\alpha, 0\rangle$. Here we take all the non-linearities and rotating terms into account, so that the dynamics is more involved. First of all, in addition to the trajectories (in blue) converging close to $|\pm\alpha, 0\rangle$, some trajectories (in red) converge towards excited limit cycles named hereafter the satellite. Due to the transcendental (sine) coupling in the Hamiltonian Eq. (4), there are infinitely many limit cycles that are solution of the dynamical system Eq. (6). The satellite presented on Fig. 3 are the first excited stable ones. To summarize, we got two sets of limit cycles: the blue ones corresponding to the cat states and the red satellites. The question becomes now: how do these satellites impact the dynamics that generates the blue limit-cycles.

B. Study of stability of the limit-cycles : Hill's method

In order to answer this question, we study the stability of the satellites thanks to Hill's method and follow the steps described in [9]. We first apply harmonic balance and find an approximation of the satellite limit cycle (yellow dashed lines on Fig. 3): the error on the approximation is controlled by the number of harmonics that are kept in the Ansatz for the harmonic balance, the larger, the more precise. Note that you can also derive analytically the center and the radii of the satellite Appendix B.

To study the stability of the satellite limit-cycle, we focus on the dynamics at the vicinity of them, like we would do for a fixed point. Let $\boldsymbol{x}^*(t)$ be the periodic solution of a limit-cycle, we perturb this solution

$$\boldsymbol{x}(t) = \boldsymbol{x}^*(t) + e^{\mu t}\boldsymbol{\delta x}(t). \tag{7}$$

Since we work in a 4d space, they are 4 complex numbers describing the dynamics of the perturbation $\left\{e^{\mu_i T}\right\}_{i\in\{1,2,3,4\}}$: they are called the Floquet multipliers[10] There are three scenarii to take into considerations

- $\forall i\, |e^{\mu_i T}| < 1$: nearby trajectories spiral towards the limit cycle, it is stable
- $\exists i\, |e^{\mu_i T}| > 1$:nearby trajectories spiral away of the limit cycle, it is unstable
- $\exists i\, e^{\mu_i T} = -1$:the limit cycle experience period doubling, it becomes unstable but a new limit-cycle with a period twice as big appears and its stability should be investigated

The first parameter we tune is the two-photon exchange rate g_2. For low value of g_2, the limit cycle is stable, then it undergoes period-doubling as g_2 is increased and becomes unstable for large value. A new cycle has appeared due to the period-doubling, which is stable at first but quickly becomes unstable as well. The period-2 orbit actually loses stability via a Neimark–Sacker bifurcation (complex pair crossing $|e^{\mu_i T}| = 1$). [TMD: Link NS bifurcation and chaos in previous works?] We now show that this instability threshold is related to the onset of chaos in this system.

C. Characterizing chaos

The most accessible signature of chaos is to analyze phase space trajectories as done on Figs. 5a to 5c. For low value of g_2, all trajectories in phase space are integrable (cf Fig. 5a), then when g_2 exceeds the stability threshold, two chaotic basins develop around the satellites (cf Fig. 5b) and finally, if g_2 keeps increasing, these basins grow and take over the phase space (cf Fig. 5c).

In order to properly characterize the classical chaos, we use the maximal Lyapunov exponent. We run into the issue that a Lyapunov exponent is defined for a given trajectory in phase space, *i.e.* a given initial point. We chose to focus on the trajectory initialized in the phase-space origin $(0, 0)$. The reason is that a cat state is roughly composed of two features: two lobes and fringes in between them. The two lobes corresponds to attractors in phase space, so we expect the fringes to be more impacted by the chaotic basins. The fringes are centered on the origin, so we chose to focus on trajectories starting on this point.

On Fig. 5d, we tune two parameters: the two-photon exchange rate g_2 on the x axis and the buffer drive ε_d, which determine the cat size (when a cat can be defined). This heatmap can be separated in three blocs:

- The blue region on the left (with point a), where the trajectory initialized at the origin is regular
- The blue region in the middle (with point b, the chaotic basin develop around the satellite limit-cyles but the trajectory that starts in the origin stay regular

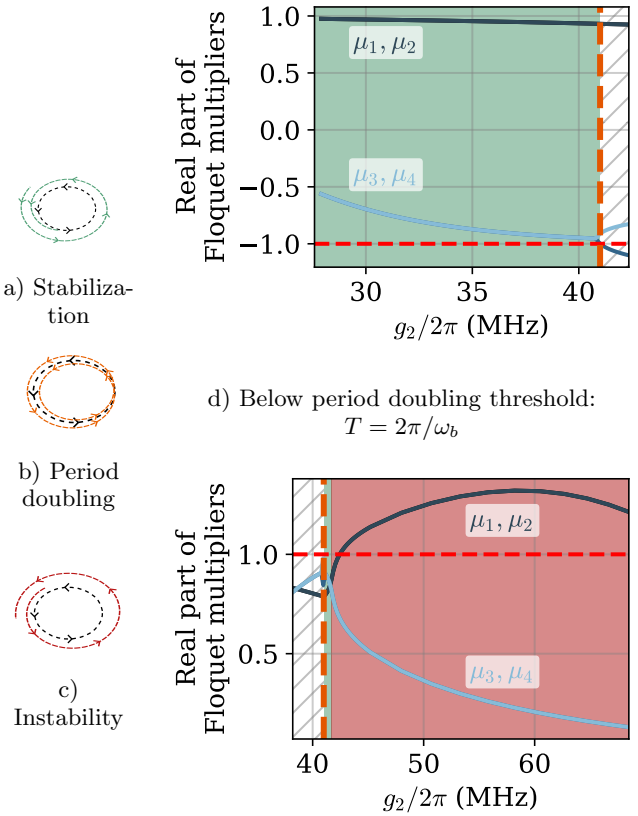
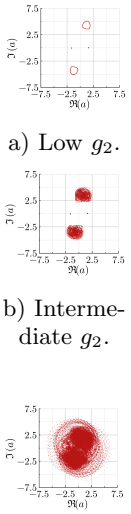


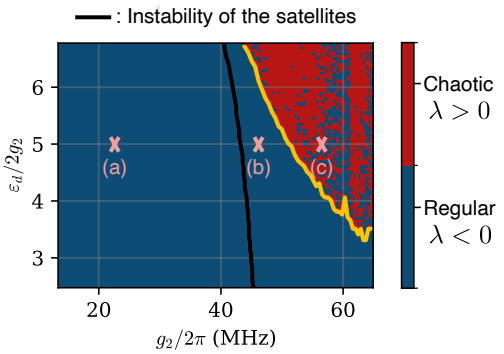
FIG. 4: Study of the stability of the satellite limit-cycles.



a) Low g_2.

b) Intermediate g_2.

c) High g_2.



d) Heatmap of the maximum Lyapunov exponent for the trajectory starting in the phase space origin ($a = 0, b = 0$). The yellow line is just there to separate the regular region (blue) from the chaotic one (red)

FIG. 5: Classical onset of chaos

- The (vastly) red region on the right (with point c), the trajectory is (almost) always chaotic [11].

We have found an upper threshold in some classical limit of the autoparametric dissipative cat, can we do the same on the quantum model directly?

IV. QUANTUM ANALYSIS

A difficulty that arises in the quantum model is to find the right metric for quantum chaos. Quantum chaos in open quantum systems has been investigated under diverse lenses: inverse participation ratio [12], observation entropy [13], truncated Wigner method [14], spectral properties of the Lindbladian [15]. In order to be able to compare the quantum results with the classical prediction, we chose to work with a metric that provides a quantum mechanical analog to the maximal Lyapunov exponent: the Out-Of-Time-Order Correlator (OTOC) [16]. The OTOC is defined as

$$C(t) = -\left\langle [\hat{x}_a(t), \hat{p}_a(0)]^2 \right\rangle \tag{8}$$

and at short time, $C(t) \sim e^{2\lambda t}$, where λ is the maximal Lyapunov exponent when we take the quasiclassical limit of the commutator [17].

We propagate the position operator over time with the adjoint Lindblad master equation [18, p. 125]

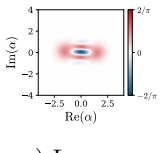
$$\dot{\hat{x}}_a(t) = i[H(t), \hat{x}_a(t)] + \kappa_b \left(\hat{b}^\dagger \hat{x}_a(t) \hat{b} - \frac{1}{2}\{\hat{b}^\dagger \hat{b}, \hat{x}_a(t)\} \right), \tag{9}$$

, compute the OTOC over an approximation of the stabilized cat-state and fit the OTOC as an exponential.

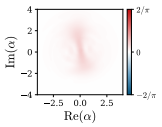
First, we see two regimes: at low values of the two-photon exchange rate g_2, the cat is stabilized (cf Fig. 6a) and for large values of g_2, the Wigner distribution is delocalized over the phase space (cf Fig. 6b). The evolution of the OTOC is showed in the inset of Fig. 6: for small g_2, the OTOC stays flat and for large g_2, it grows like an exponential with some modulation (since the Lindbladian is time-dependent[19]).

Finally, we draw heatmap of the Lyapunov exponent extracted from the OTOC where we again act on the two-photon exchange rate g_2 and the buffer drive ε_d. We again find that increasing either parameter drive the system from a regular dynamics to chaos. Nevertheless, the threshold in the quantum model is below the classical threshold: Lyapunov exponent from the OTOC are positive even below the yellow line on Fig. 6. This is explained by the intrinsic quantum delocalization: classical dynamics can be oblivious to the chaotic basin whereas the quantum dynamics can be more easily impacted by them.

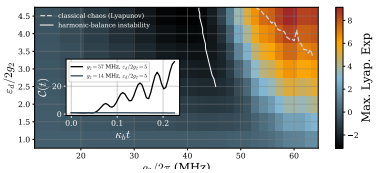
One questions remains though: are there ways to push the threshold to larger parameters range?



a) Low g_2.



b) High g_2.



c) Heatmap of the maximum Lyapunov exponent found by fitting the OTOC. The average is taken on a cat-state.

FIG. 6: Quantum onset of chaos. The subfigures Fig. 6a and Fig. 6b represent Wigner distribution obtained after some times when the initial state is a cat state.

V. MITIGATION TECHNIQUES

In this last section, we present three ways to keep the regular dynamics on larger range.

A. Increasing the single photon loss

A first solution, which allows one to keep the initial circuit, would be to increase the single photon loss rate κ_b on the buffer. The instability threshold increases linearly with the single photon rate [TMD: Explanation?], at large value of the rate, cf Fig. 7.

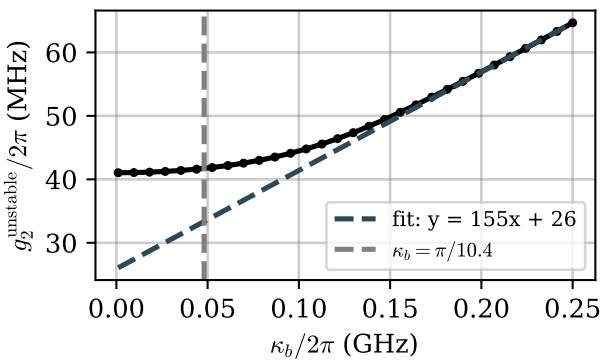


FIG. 7: Study of the instability threshold when the single photon rate on the buffer is varied. The gray dashed line corresponds to the value taken for the study in hereinabove plots.

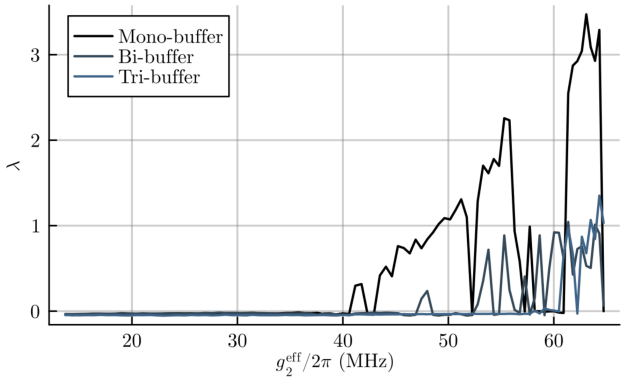


FIG. 8: Maximum Lyapunov exponent λ for the trajectory initiated in the phase-space origin

B. Multi-buffer architecture

In a similar spirit, instead of increasing the single photon loss rate, we could maybe introduce new loss channels to mitigate the apparition of chaos? The second solution is to increase the number of buffers, so that the energy is split between them. The Hamiltonian of such a system reads

$$H_{\text{m.b.}} = \omega_a \hat{a}^\dagger \hat{a} + \sum \Big\{ \omega_{b_n} \hat{b}_n^\dagger \hat{b}_n + \varepsilon_{d_n} \cos(\omega_{d_n} t)(\hat{b}_n + \hat{b}_n^\dagger) - g_n \left[\sin\left(\hat{\varphi}_a + \hat{\varphi}_{b_n}\right) - \hat{\varphi}_a - \hat{\varphi}_{b_n}\right] \Big\}. \tag{10}$$

Shall we perform RWA on this system, we would obtain

$$H_{\text{m.b.}}^{\text{RWA}} = \sum \varepsilon_{d_n} (\hat{b}_n + \hat{b}_n^\dagger)/2 + \left(a^2 \sum_n g_{n,2} b_n^\dagger + \text{h.c.} \right), \tag{11}$$

hence we can define some effective two-photon exchange rate

$$g_2^{\text{eff}} = \sum_n g_{n,2}. \tag{12}$$

We compute the maximal Lyapunov for one, two and three buffers which supposedly have the same frequency $\omega_{b_n} = 2\omega_a$ and identical drive amplitude. As shown on Fig. 8, the more buffer we add, the higher the threshold for chaos.

An example of a circuit that would implement the multi-buffer Hamiltonian Eq. (10) is presented on Fig. 9. However such circuit can be quite painful to design and work with since you increase the number of modes to fabricate and to calibrate afterwards. Couldn't we find another solution?

C. Parity-protected cat qubit

Should the recommendation from this article be condensed in one sentence, this would likely be: if you seek for less chaos, remove as many harmonics in your Hamiltonian as you can. This is what we demonstrate here: we focus on the design presented on [TMD: Cite Antoine Marquet?] whose Hamiltonian reads

$$H_{\text{p.p.}} = \omega_a a^\dagger a + \omega_b b^\dagger b + \varepsilon_d \cos(\omega_d t)(\hat{b} + \hat{b}^\dagger) - g \left[\cos\left(\hat{\varphi}_a\right) \sin\left(\hat{\varphi}_b\right) - \hat{\varphi}_b\right]. \tag{13}$$

We call this the parity-protected version of our initial Hamiltonian Eq. (4): the sine coupling is replaced with this product of cosine and sine, such that, under the change $\hat{a} \to -\hat{a}$, $H_{\text{p.p.}}$ stays the same, just as with H^{RWA}.

This Hamiltonian has also satellite, as shown on Fig. 10a, but they don't undergo any period doubling. Indeed, none of the Floquet multiplier never reaches -1 as shown on Fig. 10b.

A circuit that implements such parity-protected Hamiltonian is drawn on Fig. 11: its mirrored architecture allows to keep parity symmetry safe, the derivation of the Hamiltonian is presented in Appendix H.

VI. CONCLUSION

In this paper we showcase a pipeline for the study of chaos in Floquet quantum systems: first go to the classical limits, compute the instability threshold with Hill method, characterize the onset of chaos with the maximal Lyapunov exponent and measure the OTOC in the quantum model. We applied this workflow on a cat qubit, namely the autoparametric dissipative cat qubit and show how such a system can be driven from a regular dynamics into a chaotic one. Finally we present three ways to mitigate this chaotic behavior: increasing the single photon loss rate, adding more loss channels,

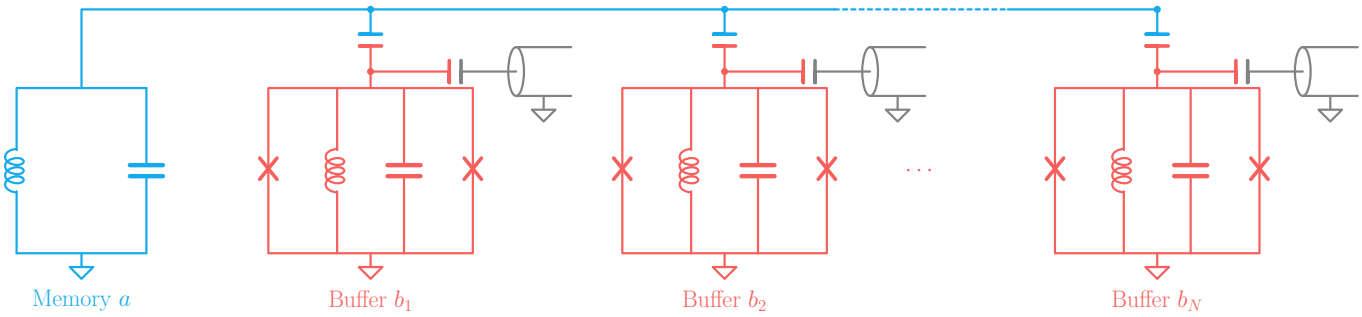
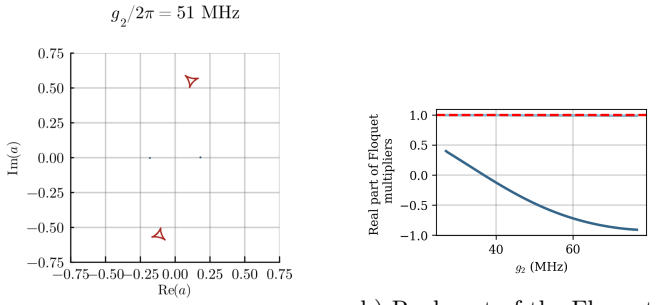


FIG. 9: Example of a multi-buffer circuit implementing $H_{\text{m.b.}}$.



a) Phase-space limit cycles

b) Real part of the Floquet multipliers

FIG. 10: Analysis of the parity protected cat qubit

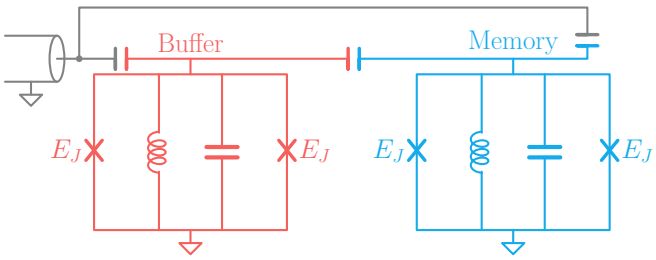


FIG. 11: Parity-protected cat qubit.

rethinking the circuit architecture to restore parity symmetry.

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Appendix A: Derivation of the classical equations of motion

In this Appendix, we derive Eq. (6) from the GKSL equation

$$\dot{\hat{\rho}} = -i[H, \hat{\rho}] + \kappa_b \mathcal{D}[\hat{b}]\hat{\rho}. \tag{A1}$$

We write $\hat{x} \equiv \hat{\varphi}_a + \hat{\varphi}_b$, with $\hat{\varphi}_a = \varphi_a(\hat{a} + \hat{a}^\dagger)$ and $\hat{\varphi}_b = \varphi_b(\hat{b} + \hat{b}^\dagger)$. The Hamiltonian splits into a linear and a nonlinear part, $H = H_l + H_{nl}$, with

$$\begin{aligned} H_l &= \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} + g\,\hat{x} \\ &\quad + \varepsilon_d \cos(\omega_d t)(\hat{b} + \hat{b}^\dagger), \qquad \text{(A2)} \\ H_{nl} &= -g \sin \hat{x}. \qquad \text{(A3)} \end{aligned}$$

1. Husimi function and correspondence rules

We work with the two-mode Husimi function

$$Q(\alpha, \beta) = \frac{1}{\pi^2} \langle \alpha, \beta | \, \hat{\rho} \, | \alpha, \beta \rangle \,, \tag{A4}$$

treating $\alpha, \alpha^*, \beta, \beta^*$ as independent variables. Its moments are anti-normally ordered expectation values, $\int d^2\alpha \, d^2\beta \; \alpha^m (\alpha^*)^n Q = \langle \hat{a}^m (\hat{a}^\dagger)^n \rangle$. Products of $\hat{\rho}$ with mode operators translate into differential operators acting on Q [TMD: Add citation]:

$$\begin{aligned} \hat{a}\hat{\rho} &\mapsto (\alpha + \partial_{\alpha^*})Q, & \hat{\rho}\hat{a}^\dagger &\mapsto (\alpha^* + \partial_\alpha)Q, \\ \hat{a}^\dagger\hat{\rho} &\mapsto \alpha^* Q, & \hat{\rho}\hat{a} &\mapsto \alpha Q, \end{aligned} \tag{A5}$$

and similarly for $\hat{b}$ with β. Successive left multiplications compose in order, $\hat{O}_1\hat{O}_2\hat{\rho} \mapsto O_1^L O_2^L Q$, and right multiplications in reverse order. Below, $\partial_\alpha(AQ) + \text{c.c.}$ is shorthand for $\partial_\alpha(AQ) + \partial_{\alpha^*}(A^*Q)$.

2. Linear part

For a single mode with $H = \omega \hat{a}^\dagger \hat{a} + F(t)(\hat{a} + \hat{a}^\dagger)$, F real, Eq. (A5) gives

$$\begin{aligned} -i[\omega \hat{a}^\dagger \hat{a}, \hat{\rho}] &\mapsto -i\omega\big[\alpha^* \partial_{\alpha^*} - \alpha \partial_\alpha\big]Q \\ &= \partial_\alpha(i\omega\alpha Q) + \text{c.c.}, \qquad \text{(A6)} \\ -iF[\hat{a} + \hat{a}^\dagger, \hat{\rho}] &\mapsto -iF\big(\partial_{\alpha^*} - \partial_\alpha\big)Q \\ &= \partial_\alpha(iFQ) + \text{c.c.} \qquad \text{(A7)} \end{aligned}$$

For the dissipator, the three terms of $\mathcal{D}[\hat{b}]\hat{\rho}$ map onto $(\beta + \partial_{\beta^*})(\beta^* + \partial_\beta)Q$, $-\frac{1}{2}\beta^*(\beta + \partial_{\beta^*})Q$ and $-\frac{1}{2}\beta(\beta^* + \partial_\beta)Q$,

which sum to

$$\kappa_b \mathcal{D}[\hat{b}]\hat{\rho} \mapsto \partial_\beta\Big(\frac{\kappa_b}{2}\beta Q\Big) + \text{c.c.} + \kappa_b \partial_\beta \partial_{\beta^*} Q. \tag{A8}$$

Collecting all terms, the linear part of the dynamics reads

$$\dot{Q}\big|_l = \partial_\alpha(A^\alpha Q) + \partial_\beta(A^\beta Q) + \text{c.c.} + \kappa_b \partial_\beta \partial_{\beta^*} Q, \tag{A9}$$

with the drift coefficients

$$A^\alpha = i\omega_a \alpha + ig\varphi_a, \tag{A10}$$

$$A^\beta = \Big(i\omega_b + \frac{\kappa_b}{2}\Big)\beta + i\varepsilon_d \cos(\omega_d t) + ig\varphi_b. \tag{A11}$$

3. Nonlinear part

Since left (right) multiplications compose in (reverse) order, the rules (A5) extend term by term to any power series f of $\hat{x}$:

$$f(\hat{x})\hat{\rho} \mapsto f(D_L)Q, \qquad \hat{\rho} f(\hat{x}) \mapsto f(D_R)Q, \tag{A12}$$

with the first-order differential operators

$$D_L = \varphi_a(\alpha + \alpha^* + \partial_{\alpha^*}) + \varphi_b(\beta + \beta^* + \partial_{\beta^*}), \tag{A13}$$

$$D_R = \varphi_a(\alpha + \alpha^* + \partial_{\alpha}) + \varphi_b(\beta + \beta^* + \partial_{\beta}). \tag{A14}$$

Hence $-i[H_{nl}, \hat{\rho}] \mapsto ig\,[\sin D_L - \sin D_R]\,Q$, and the full equation for the Husimi function is

$$\begin{aligned}\dot{Q} = {}& \partial_\alpha(A^\alpha Q) + \partial_\beta(A^\beta Q) + \text{c.c.} \\ & + \kappa_b \partial_\beta \partial_{\beta^*} Q + ig\,[\sin D_L - \sin D_R]\,Q.\end{aligned} \tag{A15}$$

Although not needed below, the action of D_L can be made explicit. Writing $\partial_{\alpha^*} + \alpha + \alpha^* = e^{-h}\partial_{\alpha^*} e^{h}$ with $h = \alpha\alpha^* + (\alpha^*)^2/2$, and similarly for β, the exponential e^{iD_L} reduces to a translation of α^* and β^*:

$$\begin{aligned}e^{iD_L} Q(\alpha, \alpha^*, \beta, \beta^*) = {}& e^{iX - (\varphi_a^2 + \varphi_b^2)/2} \\ & \times Q(\alpha, \alpha^* + i\varphi_a, \beta, \beta^* + i\varphi_b),\end{aligned} \tag{A16}$$

with $X = \varphi_a(\alpha + \alpha^*) + \varphi_b(\beta + \beta^*)$, and $e^{iD_R}Q$ is obtained by exchanging the roles of α and α^* (resp. β and β^*). The nonlinearity thus couples Q at points separated by $\pm i\varphi_{a,b}$: Eq. (A15) is nonlocal and reduces to a Fokker–Planck equation only after truncation at small $\varphi_{a,b}$.

4. Equations for the means

We define $\mu_a = \int d^2\alpha\, d^2\beta\ \alpha Q = \langle \hat{a} \rangle$ and $\mu_b = \langle \hat{b} \rangle$. Integrating Eq. (A9) by parts, $\int \alpha\, \partial_\alpha(A^\alpha Q) = -\langle A^\alpha \rangle$, while the other linear terms do not contribute to $\dot{\mu}_a$. For the nonlinear term, we use $\alpha f(D_L) Q = \pi^{-2} \langle \alpha, \beta | \, f(\hat{x}) \hat{\rho}\, \hat{a} \, | \alpha, \beta \rangle$, $\alpha f(D_R) Q = \pi^{-2} \langle \alpha, \beta | \, \hat{\rho} f(\hat{x}) \hat{a} \, | \alpha, \beta \rangle$, and the completeness relation $\int d^2\alpha\, d^2\beta\ |\alpha, \beta\rangle \langle \alpha, \beta| = \pi^2$:

$$\begin{aligned}\int d^2\alpha\, d^2\beta\ \alpha\,[f(D_L) - f(D_R)]\,Q &= \langle [\hat{a}, f(\hat{x})] \rangle \\ &= \varphi_a \langle f'(\hat{x}) \rangle.\end{aligned} \tag{A17}$$

With $f = \sin$, we obtain the exact (Ehrenfest) equations

$$\begin{cases} \dot{\mu}_a &= -i\omega_a \mu_a + ig\varphi_a\big(\langle \cos\hat{x} \rangle - 1\big), \\ \dot{\mu}_b &= -\Big(i\omega_b + \frac{\kappa_b}{2}\Big)\mu_b - i\varepsilon_d \cos(\omega_d t) \\ & \quad + ig\varphi_b\big(\langle \cos\hat{x} \rangle - 1\big), \end{cases}$$

where the -1 comes from the linear term $g\hat{x}$ in H_l.

To close these equations, we assume a Gaussian state, characterized by the second central moments of Q,

$$\begin{aligned} \sigma_a^2 &= \langle |\delta\alpha|^2 \rangle, \quad \tilde{\sigma}_a^2 = \langle \delta\alpha^2 \rangle, \quad c = \langle \delta\alpha\, \delta\beta \rangle, \\ \sigma_b^2 &= \langle |\delta\beta|^2 \rangle, \quad \tilde{\sigma}_b^2 = \langle \delta\beta^2 \rangle, \quad d = \langle \delta\alpha\, \delta\beta^* \rangle, \end{aligned} \tag{A18}$$

with $\delta\alpha = \alpha - \mu_a$, $\delta\beta = \beta - \mu_b$. For a Gaussian state $\langle e^{i\hat{x}} \rangle = e^{i\bar{X} - K}$, with $\bar{X} = \langle \hat{x} \rangle = \varphi_a(\mu_a + \mu_a^*) + \varphi_b(\mu_b + \mu_b^*)$ and $K = \mathrm{Var}(\hat{x})/2$. Because the moments of Q are anti-normally ordered, $\mathrm{Var}(\hat{a} + \hat{a}^\dagger) = 2(\sigma_a^2 + \mathrm{Re}\,\tilde{\sigma}_a^2) - 1$, so that

$$\begin{aligned} K = {}& \varphi_a^2\Big(\sigma_a^2 + \mathrm{Re}\,\tilde{\sigma}_a^2 - \tfrac{1}{2}\Big) + \varphi_b^2\Big(\sigma_b^2 + \mathrm{Re}\,\tilde{\sigma}_b^2 - \tfrac{1}{2}\Big) \\ & + 2\varphi_a\varphi_b\,\mathrm{Re}(c + d), \end{aligned} \tag{A19}$$

and $\langle \cos\hat{x} \rangle = e^{-K} \cos\bar{X}$.

The classical limit consists in neglecting the fluctuations of $\hat{x}$, $K \to 0$, i.e. $\langle \cos\hat{x} \rangle \to \cos\bar{X}$. In terms of the moments of Q, this corresponds to an isotropic Gaussian with $\sigma_a^2 = \sigma_b^2 = 1/2$ and $\tilde{\sigma}_a^2 = \tilde{\sigma}_b^2 = c = d = 0$. Identifying $a \equiv \mu_a$ and $b \equiv \mu_b$, Eq. (A18) then reduce exactly to Eq. (6). Note that a coherent state has $\sigma_{a,b}^2 = 1$, hence $K = (\varphi_a^2 + \varphi_b^2)/2$: the first quantum correction renormalizes the cosine term by $e^{-(\varphi_a^2 + \varphi_b^2)/2}$.

Appendix B: Analytical derivation of the harmonic balance

Equations of motion. We consider the classical dynamics generated by

$$\begin{aligned} H(t) = {}& \omega_a |a|^2 + \omega_b |b|^2 - 2E_J \sin\varepsilon_p \big[\sin\hat{X} - \hat{X}\big] \\ & + \varepsilon_d \cos(\omega_d t)\,(b + b^*), \\ \hat{X} = {}& \varphi_a(a + a^*) + \varphi_b(b + b^*), \end{aligned} \tag{B1}$$

with $\dot{a} = -i\,\partial H/\partial a^*$ and $\dot{b} = -i\,\partial H/\partial b^* - \kappa_b b$. We take $\omega_d = \omega_b = 2\omega_a$. We rescale the fields, $\varphi_a a \to a$ and $\varphi_b b \to b$, use the dimensionless time $\tau = \omega_a t$, and move

to the frame rotating at the mode frequencies, $a \to a\,e^{i\tau}$, $b \to b\,e^{2i\tau}$. With

$$\begin{aligned} \beta'_a &= \frac{2E_J \sin\varepsilon_p\,\varphi_a^2}{\omega_a}, & \beta'_b &= \frac{2E_J \sin\varepsilon_p\,\varphi_b^2}{\omega_b}, \\ \varepsilon'_d &= \frac{\varepsilon_d \varphi_b}{\omega_b}, & \kappa'_b &= \frac{\kappa_b}{\omega_b}, \end{aligned} \tag{B2}$$

the equations of motion read

$$\begin{aligned} \frac{da}{d\tau} &= i\beta'_a\, e^{i\tau}\big[\cos X - 1\big], \\ \frac{db}{d\tau} &= 2i\beta'_b\, e^{2i\tau}\big[\cos X - 1\big] \\ &\quad - i\varepsilon'_d\big(1 + e^{4i\tau}\big) - 2\kappa'_b b, \\ X(\tau) &= a\,e^{-i\tau} + b\,e^{-2i\tau} + \text{c.c.}, \end{aligned} \tag{B3}$$

where X is the phase across the junction.

Harmonic balance. We look for periodic solutions by inserting a finite Fourier Ansatz for $a(\tau)$ and $b(\tau)$ into Eq. (B3) and equating the Fourier components retained in the Ansatz. For any Ansatz, the junction phase takes the form

$$X(\tau) = C + \sum_{j\geq 1}\big(h_j e^{-ij\tau} + h_j^* e^{ij\tau}\big), \tag{B4}$$

with $C \in \mathbb{R}$ and $h_j = |h_j| e^{i\phi_j}$. Using the Jacobi–Anger expansion $e^{iz\cos\vartheta} = \sum_k i^k J_k(z) e^{ik\vartheta}$ for each harmonic, one finds $\cos X = \sum_{N\in\mathbb{Z}} F_N e^{iN\tau}$ with

$$\begin{aligned} F_N = \sum_{\substack{\{k_j\}\in\mathbb{Z} \\ \sum_j j k_j = N}} &\Big[\prod_j J_{k_j}\big(2|h_j|\big) e^{-ik_j\phi_j}\Big] \\ &\times \cos\Big(C + \frac{\pi}{2}\sum_j k_j\Big). \end{aligned} \tag{B5}$$

Two properties will be used repeatedly. First, if the harmonics are phase locked as $h_j = r_j e^{ij\theta}$ with $r_j \in \mathbb{R}$ (signed), then $e^{-ik_j\phi_j} J_{k_j}(2|h_j|) = e^{-ijk_j\theta} J_{k_j}(2r_j)$ (if $r_j < 0$, then $\phi_j = j\theta + \pi$ and we use $(-1)^k J_k(z) = J_k(-z)$). The constraint $\sum_j j k_j = N$ then factors out the phase, $F_N = e^{-iN\theta} R_N$, with

$$R_N = \sum_{\substack{\{k_j\}\in\mathbb{Z} \\ \sum_j j k_j = N}} \Big[\prod_j J_{k_j}(2r_j)\Big] \cos\Big(C + \frac{\pi}{2}\sum_j k_j\Big) \in \mathbb{R}. \tag{B6}$$

Second, at lowest order in the phase, $\cos X - 1 \simeq -X^2/2$, so that

$$F_N \simeq \delta_{N,0} - \tfrac{1}{2}\big(X^2\big)_N, \tag{B7}$$

where $(X^2)_N$ is the $e^{iN\tau}$ component of X^2.

1. Cat states

a. Derivation

The cat states are fixed points in the rotating frame, $a = a_0 = |a_0| e^{i\theta_{a_0}}$ and $b = b_0$. Then $C = 0$, $h_1 = a_0$, $h_2 = b_0$, and we keep only the constant component of Eq. (B3), which selects F_{-1} for the memory and F_{-2} for the buffer:

$$\begin{cases} 0 = F_{-1}, \\ 0 = 2i\beta'_b F_{-2} - i\varepsilon'_d - 2\kappa'_b b_0, \end{cases} \tag{B8}$$

with

$$\begin{aligned} F_N = \sum_{m\in\mathbb{Z}} &J_{N-2m}\big(2|a_0|\big) J_m\big(2|b_0|\big) \\ &\times e^{-i(N-2m)\theta_{a_0} - im\theta_{b_0}} \cos\Big[\frac{(N-m)\pi}{2}\Big]. \end{aligned} \tag{B9}$$

We look for solutions with an empty buffer, $b_0 = 0$, so that only $m = 0$ contributes. The memory equation is then satisfied identically, since $F_{-1} \propto \cos(-\pi/2) = 0$, and $F_{-2} = -J_2(2|a_0|)e^{2i\theta_{a_0}}$. The buffer equation becomes

$$J_2\big(2|a_0|\big)\, e^{2i\theta_{a_0}} = -\frac{\varepsilon'_d}{2\beta'_b}. \tag{B10}$$

For $\varepsilon'_d > 0$ this fixes $\theta_{a_0} = \pm\pi/2$ and $J_2(2|a_0|) = \varepsilon'_d/(2\beta'_b)$, which has solutions only if $\varepsilon'_d/(2\beta'_b) \leq \max_z J_2(z) \simeq 0.49$. Using $J_2(2x) = x^2/2 - x^4/6 + O(x^6)$, the lowest order gives the two cat states

$$\begin{aligned} &a_0 = \pm i\sqrt{\varepsilon'_d/\beta'_b}, \quad \text{i.e.} \quad |\alpha|^2 = \frac{|a_0|^2}{\varphi_a^2} = \frac{\varepsilon_d}{2g_2}, \\ &g_2 = E_J \sin\varepsilon_p\,\varphi_a^2\varphi_b, \end{aligned} \tag{B11}$$

in the original units, and the next order gives $|a_0|^2 = \frac{3}{2}\big[1 - \sqrt{1 - 4\varepsilon'_d/(3\beta'_b)}\big]$.

b. Equivalent Hamiltonian

Expanding Eq. (B1) to third order in $\hat{X}$ gives $-2E_J \sin\varepsilon_p[\sin\hat{X} - \hat{X}] \simeq (E_J \sin\varepsilon_p/3)\hat{X}^3$. For $\omega_b = 2\omega_a$, the only term of $\hat{X}^3$ that is resonant in the rotating frame is $g_2(a^2 b^* + \text{c.c.})$, which yields the two-photon Hamiltonian

$$\begin{aligned} H_{\text{2ph}} = \omega_a|a|^2 + \omega_b|b|^2 + \varepsilon_d\cos(\omega_d t)(b + b^*) \\ + g_2\big(a^2 b^* + a^{*2} b\big). \end{aligned} \tag{B12}$$

With the same rescaling as above and after dropping the counter-rotating drive term $e^{4i\tau}$, its equations of motion read

$$\begin{cases} \dfrac{da}{d\tau} = -i\beta'_a\, a^* b, \\ \dfrac{db}{d\tau} = -i\beta'_b\, a^2 - i\varepsilon'_d - 2\kappa'_b b. \end{cases} \tag{B13}$$

The fixed points satisfy $a_0^* b_0 = 0$ and $\beta'_b a_0^2 + \varepsilon'_d - 2i\kappa'_b b_0 = 0$, so that $b_0 = 0$ and $a_0^2 = -\varepsilon'_d/\beta'_b$. This is exactly Eq. (B11). In fact, since $J_2(2|a_0|)e^{2i\theta_{a_0}} = a_0^2/2 + O(|a_0|^4)$, Eq. (B10) is the full-cosine version of the two-photon fixed-point condition: H_{2ph} reproduces the cat states at leading order in the junction phase.

2. Satellites

a. Derivation

Ansatz. The satellites are periodic orbits of Eq. (B3) with period 2π (i.e. $2\pi/\omega_a$). We use

$$\begin{cases} a(\tau) = a_0 + a_1 e^{i\tau} + a_{-1} e^{-i\tau}, \\ b(\tau) = b_0 + b_{-1} e^{-i\tau} + b_2 e^{2i\tau}. \end{cases} \tag{B14}$$

In the laboratory frame:

- a_0 and b_0 oscillate at ω_a and $\omega_b = 2\omega_a$;
- a_1 and b_2 are static displacements;
- a_{-1} is a memory component at $2\omega_a$;
- b_{-1} is a buffer component at $3\omega_a$.

This choice follows from the nonlinearity. Through the quadratic term of $\cos X - 1$, the leading harmonics a_0 and b_0 generate the frequencies $0, \omega_a, 2\omega_a, 3\omega_a, 4\omega_a$. We keep the components detuned by at most ω_a from the mode frequencies, namely the static memory displacement, a_{-1} and b_{-1}. We also keep the static buffer displacement b_2, which is driven by the same Fourier component F_0 as a_1 and contributes to the static junction phase.

The buffer component at ω_a is as close to resonance as these, but it is driven by F_{-1}, which vanishes on any solution (see below). An Ansatz containing it therefore returns a vanishing amplitude and misses the components above.

With Eq. (B14), the junction phase is

$$\begin{aligned} X(\tau) &= C + \big(a_0 e^{-i\tau} + Y e^{-2i\tau} + b_{-1} e^{-3i\tau} + \text{c.c.}\big), \\ C &= 2\,\mathrm{Re}(a_1 + b_2), \qquad Y = b_0 + a_{-1}. \end{aligned} \tag{B15}$$

The memory and buffer components at $2\omega_a$ add up in X, so that $h_1 = a_0$, $h_2 = Y$, $h_3 = b_{-1}$, and Eq. (B5) becomes

$$\begin{aligned} F_N = \sum_{\substack{n,m,p\in\mathbb{Z} \\ n+2m+3p=N}} & J_n\big(2|a_0|\big) J_m\big(2|Y|\big) J_p\big(2|b_{-1}|\big) \\ & \times e^{-i(n\theta_{a_0} + m\theta_Y + p\theta_{b_{-1}})} \\ & \times \cos\Big[C + \frac{(n+m+p)\pi}{2}\Big]. \end{aligned} \tag{B16}$$

Harmonic balance. We equate the components e^0, $e^{i\tau}$ and $e^{-i\tau}$ of the memory equation, and e^0, $e^{-i\tau}$ and $e^{2i\tau}$ of the buffer equation:

$$\begin{cases} 0 = F_{-1}, \\ a_1 = \beta'_a\big(F_0 - 1\big), \\ a_{-1} = -\beta'_a F_{-2}, \\ 0 = 2i\beta'_b F_{-2} - i\varepsilon'_d - 2\kappa'_b b_0, \\ \big(\kappa'_b - \frac{i}{2}\big)\, b_{-1} = i\beta'_b F_{-3}, \\ \big(1 - i\kappa'_b\big)\, b_2 = \beta'_b\big(F_0 - 1\big). \end{cases} \tag{B17}$$

Three consequences follow directly.

- Since X is real, F_0 is real, and therefore $a_1 \in \mathbb{R}$.
- The static components are slaved to each other, $b_2 = \beta'_b a_1/[\beta'_a(1 - i\kappa'_b)]$, which gives

$$C = 2\gamma\, a_1, \qquad \gamma = 1 + \frac{\beta'_b}{\beta'_a(1 + \kappa'^2_b)}. \tag{B18}$$

- The third and fourth lines of Eq. (B17) share F_{-2}, so the drive fixes the memory component at $2\omega_a$ exactly, independently of how the cosine is truncated:

$$a_{-1} = -\frac{\beta'_a}{2\beta'_b}\big(\varepsilon'_d - 2i\kappa'_b b_0\big). \tag{B19}$$

Phase reduction. We now use $\kappa'_b \ll 1$. We set $\kappa'_b \to 0$ in the last two lines of Eq. (B17), where it only gives $O(\kappa'_b)$ corrections, and keep it in the drive equation, where it competes with ε'_d. We then look for phase-locked solutions

$$\begin{aligned} a_0 &= x\, e^{i\theta}, & b_0 &= y\, e^{2i\theta}, \\ a_{-1} &= z\, e^{2i\theta}, & b_{-1} &= w\, e^{3i\theta}, \end{aligned} \tag{B20}$$

with $x, y, z, w \in \mathbb{R}$, so that $Y = (y + z)\, e^{2i\theta} \equiv \bar{y}\, e^{2i\theta}$. Signed amplitudes also cover relative phases shifted by π, which are then decided by the equations rather than fixed by hand.

With $F_N = e^{-iN\theta} R_N$, every equation becomes real except the drive equation, whose real and imaginary parts fix θ:

$$\begin{cases} R_{-1} = 0, \\ a_1 = \beta'_a\big(R_0 - 1\big), \\ z = -\beta'_a R_{-2}, \\ 2\beta'_b R_{-2} = \varepsilon'_d \cos 2\theta, \\ \varepsilon'_d \sin 2\theta = -2\kappa'_b y, \\ w = -2\beta'_b R_{-3}, \end{cases} \tag{B21}$$

with

$$\begin{aligned} R_N = \sum_{m,p\in\mathbb{Z}} & J_{N-2m-3p}(2x)\, J_m(2\bar{y})\, J_p(2w) \\ & \times \cos\Big[2\gamma a_1 + \frac{(N - m - 2p)\pi}{2}\Big]. \end{aligned} \tag{B22}$$

Eliminating θ gives $2\beta'_b R_{-2} = \sigma\, \varepsilon_{\text{eff}}(y)$, with $\varepsilon_{\text{eff}}(y) = \sqrt{\varepsilon'^2_d - 4\kappa'^2_b y^2}$ and $\sigma = \mathrm{sign}(\cos 2\theta)$. This requires $2\kappa'_b|y| \le \varepsilon'_d$, and it gives

$$z = -\sigma\, \frac{\beta'_a}{2\beta'_b}\, \varepsilon_{\text{eff}}(y), \qquad y = \bar{y} - z. \tag{B23}$$

The two signs correspond to two families of periodic orbits:

- $\sigma = -1$, $\theta \simeq \pm\pi/2$: the cat states. Setting $a_{\pm 1} = b_2 = b_{-1} = 0$ recovers Appendix B 1 a.

- $\sigma = +1$, $\theta \simeq 0, \pi$: the satellites. They are rotated by 90° with respect to the cats, up to the dissipative tilt $\theta_{b_0} = 2\theta = -\arcsin(2\kappa'_b y/\varepsilon'_d)$, with $\theta_{a_0} = \theta_{b_0}/2$ $[\pi]$.

Equation (B21) is invariant under $(x, w) \to (-x, -w)$, i.e. under $a \to -a$, so the satellites come in pairs related by the parity of the memory.

Leading-order solution. At lowest order in the junction phase, Eq. (B7) gives

$$\begin{aligned} R_{-1} &\simeq -x(C+\bar{y}) - \bar{y}w, \\ R_0 - 1 &\simeq -\Big(\frac{C^2}{2} + x^2 + \bar{y}^2 + w^2\Big), \\ R_{-2} &\simeq -\Big(C\bar{y} + \frac{x^2}{2} + xw\Big), \\ R_{-3} &\simeq -(Cw + x\bar{y}). \end{aligned} \tag{B24}$$

For $\kappa'_b \to 0$ in γ, we have $\gamma\beta'_a = \beta'_a + \beta'_b \equiv \beta_\Sigma$, and Eq. (B21) becomes the polynomial system

$$\begin{cases} x\,(C+\bar{y}) + \bar{y}\,w = 0, \\ C = -2\beta_\Sigma\Big(\dfrac{C^2}{2} + x^2 + \bar{y}^2 + w^2\Big), \\ -2\beta'_b\Big(C\bar{y} + \dfrac{x^2}{2} + xw\Big) = \sigma\,\varepsilon_{\rm eff}(y), \\ w = 2\beta'_b\,(Cw + x\bar{y}), \end{cases} \tag{B25}$$

completed by Eq. (B23).

For $x \neq 0$, let $u = w/x$ and $\eta = 2\beta'_b\bar{y}$. The first and last lines give $C = -\bar{y}(1+u)$ and $\eta u^2 + (1+\eta)u - \eta = 0$, whose root continuously connected to $u = 0$ is

$$u(\bar{y}) = \frac{\sqrt{(1+\eta)^2 + 4\eta^2} - (1+\eta)}{2\eta} = \eta + O(\eta^2). \tag{B26}$$

The third line with $\sigma = +1$ then fixes the memory amplitude,

$$x^2 = \frac{2(1+u)\,\bar{y}^2 - \varepsilon_{\rm eff}(y)/\beta'_b}{1+2u}, \tag{B27}$$

and the second line reduces the problem to a single scalar equation for the total $2\omega_a$ amplitude $\bar{y}$:

$$\bar{y}\,(1+u) = 2\beta_\Sigma\Big[\Big(1 + \frac{(1+u)^2}{2}\Big)\bar{y}^2 + \big(1+u^2\big)\,x^2\Big]. \tag{B28}$$

Here y is related to $\bar{y}$ by $y = \bar{y} + \beta'_a\varepsilon_{\rm eff}(y)/(2\beta'_b)$ from Eq. (B23). Since $C < 0$ by the second line of Eq. (B25), the physical root has $\bar{y} > 0$ and must satisfy $x^2 > 0$.

All other amplitudes follow:

$$\begin{aligned} a_0 &= \pm x\,e^{i\theta}, & b_0 &= y\,e^{2i\theta}, \\ b_{-1} &= \pm u\,x\,e^{3i\theta}, & a_{-1} &= -\frac{\beta'_a\varepsilon_{\rm eff}(y)}{2\beta'_b}\,e^{2i\theta}, \\ a_1 &= -\frac{(1+u)\,\bar{y}}{2\gamma}, & b_2 &= \frac{\beta'_b}{\beta'_a}\,a_1. \end{aligned} \tag{B29}$$

To lowest order in η, $|b_{-1}| \simeq 2\beta'_b|a_0|\bar{y}$.

In the limit $u \to 0$ and $\kappa'_b \to 0$, Eq. (B28) reduces to $7\beta_\Sigma\bar{y}^2 - \bar{y} - 2\beta_\Sigma\varepsilon'_d/\beta'_b = 0$, i.e.

$$\begin{aligned} \bar{y} &= \frac{1 + \sqrt{1 + 56\,\beta_\Sigma^2\,\varepsilon'_d/\beta'_b}}{14\,\beta_\Sigma}, \\ |a_0|^2 &= 2\bar{y}^2 - \frac{\varepsilon'_d}{\beta'_b}, \qquad |b_0| = \bar{y} + \frac{\beta'_a\varepsilon'_d}{2\beta'_b}. \end{aligned} \tag{B30}$$

Orbits and validity. In the rotating frame, the memory follows $a_0 + a_1e^{i\tau} + a_{-1}e^{-i\tau}$, an ellipse centered on a_0 with semi-axes $|a_1| \pm |a_{-1}|$. The buffer follows $b_0 + b_2e^{2i\tau} + b_{-1}e^{-i\tau}$, which describes a trefoil, matching Fig. 3 .

Since $\bar{y} \sim 1/(7\beta_\Sigma)$, the expansion in the junction phase is quantitatively controlled only when $|a_0|, \bar{y}, |C| \ll 1$. Otherwise, the full Bessel system Eq. (B21) has to be solved numerically, using the solution of Eq. (B28) as the initial guess.

b. *Equivalent Hamiltonian*

We look for a truncated Hamiltonian whose harmonic balance with the Ansatz Eq. (B14) gives exactly the approximated equations, i.e. Eq. (B17) with F_N replaced by Eq. (B7). This is the case for the cubic expansion of Eq. (B1), $-2E_J\sin\varepsilon_p[\sin\hat{X} - \hat{X}] \to (E_J\sin\varepsilon_p/3)\hat{X}^3$, because its equations of motion are Eq. (B3) with $\cos X - 1 \to -X^2/2$. In normal-ordered monomials it reads

$$\begin{aligned} H_{\rm trunc} = {}& \omega_a|a|^2 + \omega_b|b|^2 + \varepsilon_d\cos(\omega_d t)(b + b^*) \\ &+ E_J\sin\varepsilon_p\Big[\varphi_a^2\varphi_b\,a^2b^* + \varphi_a^3\,a^2a^* + \varphi_b^3\,b\,b^{*2} \\ &\quad + \varphi_a\varphi_b^2\,a\,b^{*2} + 2\varphi_a^2\varphi_b\,|a|^2b \\ &\quad + 2\varphi_a\varphi_b^2\,a\,|b|^2 + \text{c.c.}\Big] \\ &+ E_J\sin\varepsilon_p\Big[\frac{\varphi_a^3}{3}\,a^3 + \varphi_a^2\varphi_b\,a^2b + \varphi_a\varphi_b^2\,a\,b^2 \\ &\quad + \frac{\varphi_b^3}{3}\,b^3 + \text{c.c.}\Big]. \end{aligned} \tag{B31}$$

Inserting Eq. (B14) and projecting as in Eq. (B17), its

harmonic balance reads, before phase reduction,

$$\begin{aligned}
0 &= Ca_0 + Ya_0^* + Y^* b_{-1},\\
a_1 &= -\beta_a'\Big(\frac{C^2}{2} + |a_0|^2 + |Y|^2 + |b_{-1}|^2\Big),\\
a_{-1} &= \beta_a'\Big(CY + \frac{a_0^2}{2} + a_0^* b_{-1}\Big),\\
0 &= -i\beta_b'\Big(CY + \frac{a_0^2}{2} + a_0^* b_{-1}\Big)\\
&\quad - \frac{i\varepsilon_d'}{2} - \kappa_b' b_0,\\
\big(\kappa_b' - \tfrac{i}{2}\big)\, b_{-1} &= -i\beta_b'\big(Cb_{-1} + a_0 Y\big),\\
\big(1 - i\kappa_b'\big)\, b_2 &= -\beta_b'\Big(\frac{C^2}{2} + |a_0|^2 + |Y|^2 + |b_{-1}|^2\Big),
\end{aligned} \tag{B32}$$

with $C = 2\mathrm{Re}(a_1 + b_2)$ and $Y = b_0 + a_{-1}$. After phase locking, these equations are exactly Eqs. (B23) and (B25).

The two groups of terms in Eq. (B31) play different roles.

- The first bracket contains every term that contributes to the equations of the oscillating amplitudes a_0, a_{-1}, b_0 and b_{-1}:
  - the two-photon exchange $a^2 b^*$, which alone gives the cat states of Appendix B 1 b;
  - together with $a|b|^2$, the same term produces the three-wave mixing $a_0 Y \leftrightarrow b_{-1}$ responsible for the sideband at $3\omega_a$;
  - evaluated on the static displacements, all the terms of this bracket produce the frequency shifts Ca_0, CY and Cb_{-1}.
- The second bracket, a^3, $a^2 b$, ab^2, b^3 and their complex conjugates, contributes only to the two static equations (for a_1 and b_2), through products of a_1 and b_2. These terms are nevertheless required: together with the static parts of the first bracket, they build the complete $C^2/2$ term, which is of the same order as the other terms.

Appendix C: Period-doubled satellite limit-cycle

Appendix D: Lyapunov heatmap with initial points at the center of the cat lobes

We check that the fringes are indeed the most fragile feature of the cat qubit we work with. To show this, we compute the maximal Lyapunov exponent when the trajectory is started close to a cat limit-cycle, instead of starting at the origin of phase space like in the main text. As shown on Fig. 14, the threshold for chaotic dynamics is higher, meaning we can work with higher values of the two-photon exchange rate g_2 or buffer drive amplitude ε_d and observe regular trajectories.

[TMD: Other reason why it is important to check this?]

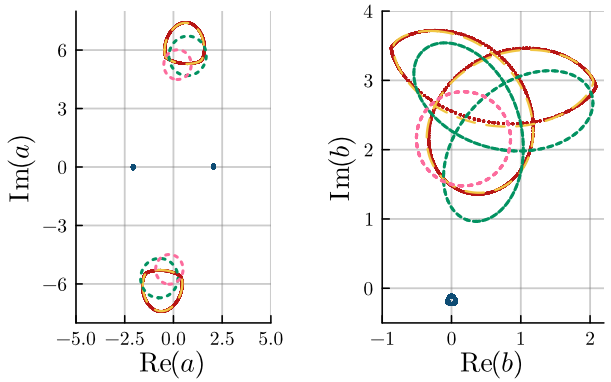


FIG. 12: Prediction of the harmonic balance. In yellow the numerical solution with many harmonics, in green the the numerical solution of the system Eq. (B29) and in pink the analytical solution Eq. (B30). Despite its simplicity, we get a good estimate of the position and size of the satellite limit cycles for the memory.

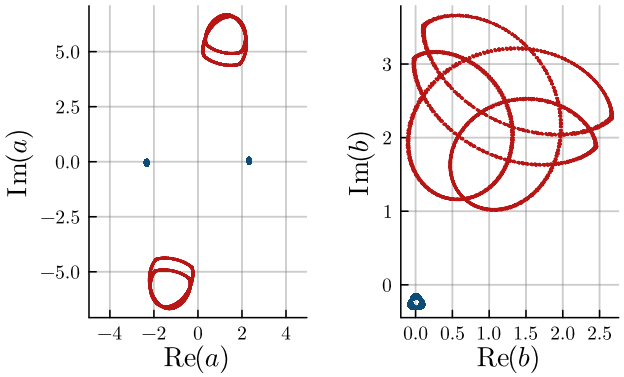


FIG. 13: Example of limit-cycles after the period-doubling

Appendix E: Lyapunov heatmap where the instability matches the blue to red transition

[TMD: Check if satellites do not exist in the light blue region, with less time for the computation of the Lyapunov exponent.]

Figure 15 shows the map for trajectories started on the satellite. Of the points where a satellite solution exists, 30.8% end on the satellite, 29.2% on a cat, and 40.0% are chaotic. From low to high g_2, the map shows four regions.

- A band about 8 MHz wide along the edge of the blank region, where the trajectory relaxes to a cat.

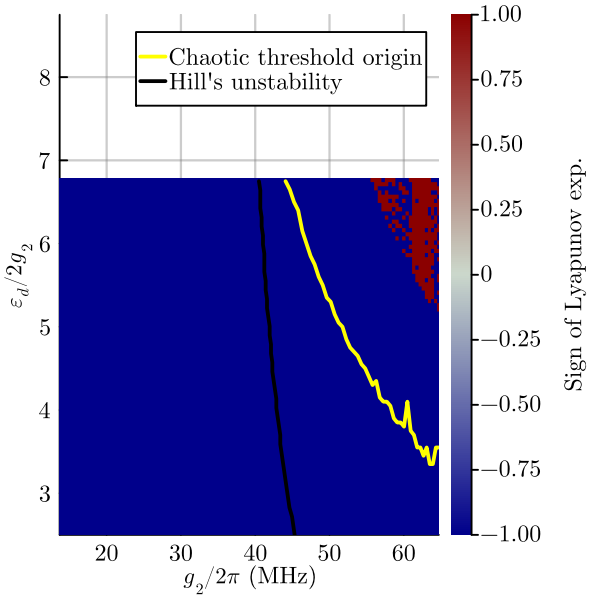


FIG. 14: Lyapunov heatmap when the trajectory is initialized close to a cat limit-cycle

There the analytic satellite exists but the actual satellite is unstable or absent, so the starting point lies in the basin of a cat.

- A band where the trajectory stays on the satellite ($\lambda < 0$, large $\langle|b|\rangle$), up to $g_2/2\pi \simeq 41$ MHz at $\varepsilon_d/2g_2 = 6.75$ and $\simeq 45$ MHz at $\varepsilon_d/2g_2 = 2.5$. Its upper edge coincides, within one or two grid cells, with the instability line of the satellite branch obtained independently from Hill's method (**??**). Along that branch, the satellite first period-doubles. At $\varepsilon_d = 10.12\, g_2$ a Floquet multiplier crosses -1 at $g_2/2\pi \simeq 41.0$ MHz. The period-two orbit then loses stability through a complex pair of multipliers leaving the unit circle (a Neimark–Sacker bifurcation) at $g_2/2\pi \simeq 41.7$ MHz. Just beyond that line, the trajectories become chaotic.

- A region of uniform chaos, up to $g_2/2\pi \simeq 50$ MHz.

- At larger g_2, a fine-grained mixture of chaotic, cat and satellite end states between neighbouring grid points.

The agreement between the onset of chaos and Hill's instability line identifies the destabilisation of the satellite branch as the route to chaos. The vacuum-start onset lies at larger g_2 than the satellite instability. This is consistent with the vacuum having to fall into the basin of the chaotic set first, rather than starting on the branch that generates it.

Limitations. The fine-grained region at large g_2 should not be read as a sharp phase diagram. There

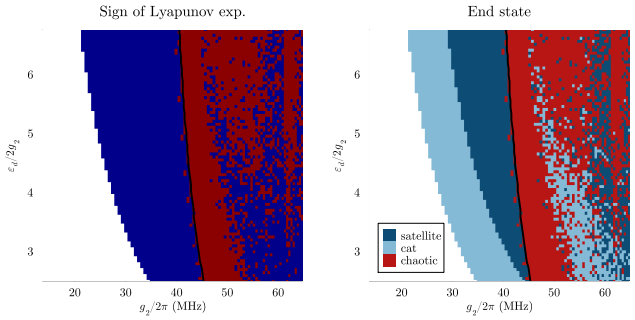


FIG. 15: Trajectories started on the satellite. Left: sign of the largest Lyapunov exponent (blue $\lambda < 0$, red $\lambda > 0$). Right: end state (satellite, cat or chaotic). Blank: no satellite solution. Black line: instability of the satellite branch from Hill's method (**??**).

the finite-time exponent is evaluated over $45/\kappa_b$, which is comparable to the lifetime of transient chaos at these drive strengths. For example, $\simeq 42/\kappa_b$ at $\varepsilon_p = 0.802$, $g_2/2\pi \simeq 50.8$ MHz (**??**). Neighbouring points can then fall on either side of the end of the transient. In the same way, the cat and satellite basins are probed with a single initial condition per grid point. The maps thus show which attractor that particular initial condition reaches within the window, not the full basin structure.

Appendix F: Computation of IPR as a signature of chaos

1. Definition

For a pure state $|\psi\rangle$ expanded on an orthonormal basis $\{|k\rangle\}$, the inverse participation ratio (IPR) is

$$\mathrm{IPR} = \sum_k |\langle k|\psi\rangle|^4. \tag{F1}$$

It equals 1 when $|\psi\rangle$ is a single basis state and $1/N$ when $|\psi\rangle$ is spread evenly over N of them. Its inverse, $N_{\mathrm{eff}} = 1/\mathrm{IPR}$, counts the basis states that effectively take part in $|\psi\rangle$. The memory is entangled with the lossy buffer, so its reduced state $\hat{\rho}_a = \mathrm{Tr}_b\, \hat{\rho}$ is mixed. We therefore use the natural generalization

$$\mathrm{IPR}(t) = \sum_{k=0}^{n_a-1} p_k(t)^2, \qquad p_k(t) = \langle k|\hat{\rho}_a(t)|k\rangle, \tag{F2}$$

where $|k\rangle$ are the Fock states of the memory. For a pure state, Eq. (F2) reduces to Eq. (F1). Only the Fock populations enter: Eq. (F2) measures how far the photon-number distribution of the memory is spread, independently of the coherences between Fock states.

2. IPR as a probe of quantum chaos

The participation ratio was introduced to quantify the localization of vibrational modes [20] and electronic wave functions [21, 22]. It became a standard measure of eigenstate delocalization in quantum chaos [23, 24]. In a regular (integrable) system, eigenstates and wave packets stay localized in a suitable unperturbed basis, and the IPR remains of order one. In a chaotic system they spread ergodically over the energetically available states, and the IPR decreases towards its random-state value, $2/(N+1)$ for a random complex vector in dimension N [23, 25, 26]. Applied to a time-evolved state instead of eigenstates, a decay of IPR(t) quantifies the dynamical spreading of an initially localized state over the basis [26]. It complements the out-of-time-order correlator, which probes operator growth.

For a cat qubit, the Fock basis of the memory is the natural reference. In the stabilized regime, the two-photon dissipation confines the memory to the cat manifold spanned by $|\pm\alpha\rangle$. Its photon-number distribution remains that of the initial cat, a Poisson distribution of mean $|\alpha|^2$ restricted to one parity. A spreading of the Fock distribution beyond it, i.e. a decrease of the IPR, signals that the dynamics leaves the cat manifold.

3. Results

Figure 16 shows IPR(T) over the $(\varepsilon_p, \varepsilon_d/g_2)$ plane. Most of the vertical gradient comes from the initial state: a larger drive prepares a larger cat, whose Fock distribution is wider, and IPR(0) decreases from 0.526 at $r = 2$ to 0.270 at $r = 9$. The dynamics is read from the departure of IPR(T) from IPR(0) along each row.

At weak pump amplitude the Fock distribution is frozen. At $\varepsilon_p = 0.2$, IPR(T)/IPR(0) ≥ 0.96 for every drive strength. This is the expected behavior of a stabilized cat qubit, whose state remains in the cat manifold.

As ε_p increases, IPR(T) decreases for every $r \geq 4$, and the decrease is larger at stronger drive. It proceeds in two stages. Up to $\varepsilon_p \approx 0.8$, IPR($T$) decreases smoothly, by at most $\sim 25\%$ relative to its value at $\varepsilon_p = 0.2$. Above $\varepsilon_p \approx 0.85$ it drops faster. At $\varepsilon_p = 1.08$ it reaches IPR(T)/IPR(0) $= 0.74$ for $r = 4$ and 0.38 for $r = 9$. In that corner, the memory effectively occupies $N_{\text{eff}} = 1/\text{IPR} \approx 9.7$ Fock states, against 3.7 for the initial cat. This is the strong spreading of the photon-number distribution expected from chaotic dynamics, which mixes the cat with Fock states outside its parity sector and its Poisson envelope. The small non-monotonic features along the rows, e.g. near $\varepsilon_p \approx 0.93$, are expected from a snapshot at a single time of a state that is still evolving.

At the smallest cat sizes, the behavior is different. For $r = 2$ ($|\alpha|^2 = 1$), IPR(T) instead increases with ε_p, by up to 7%: the Fock distribution concentrates rather than spreads. For $r = 3$, the IPR shows only a shallow minimum, 8% below IPR(0), around $\varepsilon_p \approx 0.8$. The spreading therefore requires both a large pump amplitude and a sufficiently large cat.

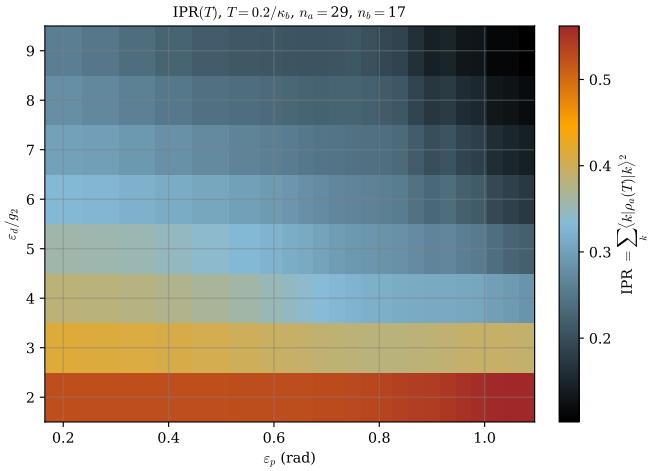


FIG. 16: Inverse participation ratio $\text{IPR}(T) = \sum_k \langle k|\hat{\rho}_a(T)|k\rangle^2$ of the memory in its Fock basis, at $T = 0.2/\kappa_b$, as a function of the pump amplitude ε_p and the drive strength ε_d/g_2. The memory starts in the even cat state of amplitude $\alpha = \sqrt{\varepsilon_d/2g_2}$. Truncations $n_a = 29$, $n_b = 17$.

The same simulations give the Wehrl entropy $S_W(T)$ of the memory. Over the 184 points of the grid, $S_W(T)$ and $-\ln \text{IPR}(T)$ are strongly correlated (Pearson coefficient 0.98). Delocalization in the Fock basis and spreading of the Husimi distribution in phase space thus tell the same story.

Two caveats apply. First, the IPR is basis dependent: a decrease signals spreading in the Fock basis of the memory, which can also come from a change of the mean photon number or from the entanglement with, and the loss through, the buffer. Its interpretation as a chaos indicator rests on the comparison with the stabilized regime, where these effects are present but leave the IPR unchanged. Second, $T = 0.2/\kappa_b$ is short compared with the buffer lifetime. Figure 16 therefore probes the onset of the spreading, not its long-time saturation.

Appendix G: OTOC but computed on a coherent state and not a cat

One could assert that the computation of the OTOC is dependent on the state it gets averaged on. Furthermore, we showed the results for a cat state, a bimodally distributed state and could be different for a more standard unimodally distributed state such as a coherent state. In this appendix, we present on Fig. 17 the maximal Lyapunov exponent derived from the OTOC when averaged on the coherent state $|\alpha\rangle$, building block of the cat state. We see no major changes.

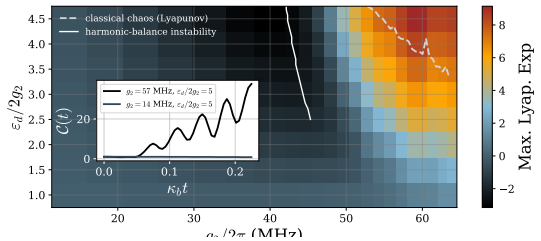


FIG. 17: OTOC averaged on a coherent state (lobe of the cat state)

Appendix H: Derivation of the parity-protected Hamiltonian

In this appendix we derive the Hamiltonian of the mirror-symmetric circuit whose nonlinearity is even in the memory quadratures,

$$-4E_J \sin[\epsilon(t)] \cos(\varphi_a \hat{x}_a) \sin(\varphi_b \hat{x}_b), \tag{H1}$$

by combining two identical asymmetrically threaded SQUIDs (ATS). The circuit is treated classically and quantized only at the end (Sec. H 4). We use the reduced flux quantum $\phi_0 = \hbar/2e$, write phases as $\phi = \Phi/\phi_0$, and define $E_L = \phi_0^2/L$ and $E_C = e^2/2C$.

1. ATS element

Each element $i = 1, 2$ consists of two junctions (E_J, capacitance C_J) and a central inductor L in parallel between an island and ground, with sub-nodes $\Phi_{i,1}$, $\Phi_{i,2}$ (inductor) and $\Phi_{i,3}$. The loop fluxes impose $\Phi_{i,2} - \Phi_{i,1} = \Phi_{l,i}(t)$ and $\Phi_{i,3} - \Phi_{i,2} = \Phi_{r,i}(t)$. Defining $\Phi_{\Sigma,i} = (\Phi_{l,i} + \Phi_{r,i})/2$, $\Phi_{\Delta,i} = (\Phi_{r,i} - \Phi_{l,i})/2$ and the island coordinate $\Phi_i = (\Phi_{i,1} + \Phi_{i,3})/2$, one has $\Phi_{i,1,3} = \Phi_i \mp \Phi_{\Sigma,i}$ and $\Phi_{i,2} = \Phi_i - \Phi_{\Delta,i}$. Assuming equal junction capacitances and that the external circuit attaches to the inductor sub-node $\Phi_{i,2}$, the element Lagrangian is $\mathcal{L}_{\mathrm{ATS},i} = C_J \dot{\Phi}_i^2 - U_i(\phi_i, t)$, up to a term depending only on the external fluxes, with

$$U_i(\phi_i, t) = \frac{E_L}{2} (\phi_i - \phi_{\Delta,i})^2 - 2E_J \cos\phi_{\Sigma,i}(t) \cos\phi_i. \tag{H2}$$

The Σ-flux generates no charge drive on Φ_i, whereas the Δ-flux enters both the inductive potential and, because the external circuit is attached at $\Phi_i - \Phi_{\Delta,i}(t)$, the kinetic terms of the coupling elements. These terms are kept explicitly, which is equivalent to the irrotational construction [27]. Unequal junction capacitances only add identical offsets to $\phi_{\Sigma,i}$ in both elements, which affect the symmetric coordinate alone and leave the parity argument unchanged.

2. Circuit Lagrangian and bias point

The inductor nodes carry a shunt capacitance C_s, are coupled to each other by C_c (and optionally L_c), and each couples through C_p to the port node at voltage $V_p(t)$. With identical Δ-fluxes $\Phi_{\Delta,1} = \Phi_{\Delta,2} = \Phi_\Delta(t)$, integrating by parts the cross terms gives

$$\begin{aligned}\mathcal{L} = \sum_{i=1,2} \Big[&\frac{C}{2}\dot{\Phi}_i^2 + \frac{C_p}{2}\big(\dot{\Phi}_i - V_p\big)^2 \\ &+ C_{\mathrm{ext}}\, \Phi_i \ddot{\Phi}_\Delta - U_i(\phi_i, t)\Big] \\ &+ \frac{C_c}{2}\big(\dot{\Phi}_1 - \dot{\Phi}_2\big)^2 - \frac{(\Phi_1 - \Phi_2)^2}{2L_c},\end{aligned} \tag{H3}$$

with $C = C_s + 2C_J$ and $C_{\mathrm{ext}} = C_s + C_p$, up to total derivatives and functions of time. The term $C_{\mathrm{ext}}\Phi_i\ddot{\Phi}_\Delta$ is the charge drive induced by the pumped Δ-flux; it is symmetric in i.

In configuration A both elements are biased identically, $\phi_{\Sigma,i} = -\pi/2 + \epsilon(t)$ and $\phi_{\Delta,i} = -\pi/2 + \eta(t)$, with $\epsilon(t) = \epsilon_p \sin(\omega_p t)$ and $\eta(t) = \eta_p \sin(\omega_p t)$. Shifting $\vartheta_i = \phi_i + \pi/2$ and absorbing the charge drive, each element contributes $C\dot{\Phi}_i^2/2 - U_i^{\mathrm{eff}}$ with

$$U_i^{\mathrm{eff}} = \frac{E_L}{2}\vartheta_i^2 - E_L^\eta\, \eta(t)\, \vartheta_i - 2E_J \sin[\epsilon(t)] \sin\vartheta_i, \tag{H4}$$

with $E_L^\eta = E_L(1 - \omega_p^2/\omega_L^2)$ and $\omega_L^2 = 1/(LC_{\mathrm{ext}})$. The factor $(1 - \omega_p^2/\omega_L^2)$ is a small charge-drive correction ($\simeq 0.98$ for typical parameters). The point $\vartheta_i = 0$ is the static equilibrium; at $\epsilon = 0$ the junctions do not contribute to the linear inductance. Equation (H4) is the classical, single-element counterpart of the ATS potential of Ref. [7].

3. Normal modes and classical Hamiltonian

We introduce the symmetric and antisymmetric coordinates $\Psi_\pm = \phi_0(\vartheta_1 \pm \vartheta_2)/2$. Using $2\cos\theta_a \sin\theta_b = \sin(\theta_b + \theta_a) + \sin(\theta_b - \theta_a)$, the Lagrangian separates into

$$\begin{aligned}\mathcal{L} = &\frac{C_+}{2}\dot{\Psi}_+^2 - \frac{\Psi_+^2}{2L_+} + \frac{C_-}{2}\dot{\Psi}_-^2 - \frac{\Psi_-^2}{2L_-} \\ &- 2C_p V_p \dot{\Psi}_+ + 2E_L^\eta\, \eta(t) \frac{\Psi_+}{\phi_0} \\ &+ 4E_J \sin[\epsilon(t)] \cos\frac{\Psi_-}{\phi_0} \sin\frac{\Psi_+}{\phi_0},\end{aligned} \tag{H5}$$

with

$$\begin{aligned} C_+ &= 2\bar{C}, & L_+ &= \frac{L}{2}, \\ C_- &= 2(\bar{C} + 2C_c), & \frac{1}{L_-} &= \frac{2}{L} + \frac{4}{L_c}, \end{aligned} \tag{H6}$$

and $\bar{C} = C + C_p$. Mirror symmetry forbids any $\Psi_+\Psi_-$ or $\dot{\Psi}_+\dot{\Psi}_-$ cross term, so $\Psi_\pm$ are exactly the normal modes.

Because the vector $(1,1)$ is an eigenvector of the capacitance matrix and the Δ-fluxes are identical, both the port and the Δ-pump couple only to Ψ_+.

With conjugate charges $Q_+ = C_+\dot{\Psi}_+ - 2C_pV_p$ and $Q_- = C_-\dot{\Psi}_-$ (i.e. $Q_\pm = Q_1 \pm Q_2$), the Legendre transform gives

$$\begin{aligned}\mathcal{H} = &\frac{Q_+^2}{2C_+} + \frac{\Psi_+^2}{2L_+} + \frac{Q_-^2}{2C_-} + \frac{\Psi_-^2}{2L_-} \\ &+ \frac{C_p}{\bar{C}}V_p(t)\,Q_+ - 2E_L^\eta\,\eta(t)\frac{\Psi_+}{\phi_0} \\ &- 4E_J\sin[\epsilon(t)]\cos\frac{\Psi_-}{\phi_0}\sin\frac{\Psi_+}{\phi_0}.\end{aligned} \tag{H7}$$

The linear frequencies are

$$\begin{aligned}\omega_b &= \frac{1}{\sqrt{L_+C_+}} = \frac{1}{\sqrt{L\bar{C}}}, \\ \omega_a &= \frac{1}{\sqrt{L_-C_-}} = \sqrt{\frac{1/L + 2/L_c}{\bar{C} + 2C_c}}.\end{aligned} \tag{H8}$$

The antisymmetric mode Ψ_- is the memory a and the symmetric mode Ψ_+ is the buffer b. Without L_c, the coupling capacitance yields $\omega_a < \omega_b$; adding L_c raises ω_a.

a. Cancellation of the linear buffer drive. The nonlinear term contains the linear piece $-4E_J\sin[\epsilon(t)]\Psi_+/\phi_0$. Using $\sin\epsilon(t) = 2\sum_{k>0,\,\text{odd}} J_k(\epsilon_p)\sin(k\omega_p t)$, the total linear drive on the buffer at ω_p is $-[2E_L^\eta\eta_p + 8E_JJ_1(\epsilon_p)]\sin(\omega_p t)\Psi_+/\phi_0$, which vanishes for

$$\eta_p = -\frac{4E_JJ_1(\epsilon_p)}{E_L^\eta} \xrightarrow{\epsilon_p \ll 1} \frac{\eta_p}{\epsilon_p} = -\frac{2E_J}{E_L^\eta}, \tag{H9}$$

the two-element counterpart of the single-ATS $g_{1,1} = 0$ condition of Ref. [7]. The residual linear drive is the third harmonic, $-8E_JJ_3(\epsilon_p)\sin(3\omega_p t)\Psi_+/\phi_0$, smaller by $J_3/J_1 \simeq \epsilon_p^2/24$. This removes the circulating pump photons in the buffer, and it does so with a parity-even drive.

b. Configuration B. Alternatively, the elements can be pumped in antiphase, $\phi_{\Sigma,1,2} = -\pi/2 \pm \epsilon(t)$, with the same static bias. The nonlinearity then becomes $-4E_J\sin\epsilon\,\cos(\Psi_+/\phi_0)\sin(\Psi_-/\phi_0)$, so the roles of the modes are exchanged: the memory must be Ψ_+ and the buffer Ψ_-, and the port must couple to Q_- only (e.g. through a balun). Because of this port requirement, configuration A is preferred.

4. Quantization

We promote $[\hat{\Psi}_\pm, \hat{Q}_\pm] = i\hbar$ and write the line voltage as $V_p(t) = V_d\cos(\omega_d t) + \hat{V}_B$, a classical buffer drive plus the transmission-line bath operator. With $\hat{\Psi}_- = \Psi_a^{\text{zpf}}\hat{x}_a$, $\hat{Q}_- = Q_a^{\text{zpf}}\hat{y}_a$ (and likewise for b with Ψ_+, Q_+), $\hat{x}_a = \hat{a} + \hat{a}^\dagger$, $\hat{y}_a = -i(\hat{a} - \hat{a}^\dagger)$ (same for b), $\Psi^{\text{zpf}} = \sqrt{\hbar Z/2}$, $Q^{\text{zpf}} = \sqrt{\hbar/2Z}$ and $Z_{a,b} = \sqrt{L_\mp/C_\mp}$, the phase zero-point fluctuations are

$$\varphi_a = \left(\frac{2E_{C_-}}{E_{L_-}}\right)^{1/4}, \qquad \varphi_b = \left(\frac{2E_{C_+}}{E_{L_+}}\right)^{1/4}, \tag{H10}$$

with $E_{C_\pm} = e^2/2C_\pm$ and $E_{L_\pm} = \phi_0^2/L_\pm$. The island phases are $\hat{\vartheta}_{1,2} = \varphi_b\hat{x}_b \pm \varphi_a\hat{x}_a$: the memory enters the two elements with opposite signs and the buffer with the same sign. Dropping zero-point energies, the system and system–bath Hamiltonians read

$$\begin{aligned}\frac{\hat{H}_s}{\hbar} &= \omega_a\hat{a}^\dagger\hat{a} + \omega_b\hat{b}^\dagger\hat{b} \\ &\quad + 2\epsilon_d\cos(\omega_d t)\,\hat{y}_b - 2\frac{E_L^\eta}{\hbar}\eta(t)\,\varphi_b\hat{x}_b \\ &\quad - 4\frac{E_J}{\hbar}\sin[\epsilon(t)]\cos(\varphi_a\hat{x}_a)\sin(\varphi_b\hat{x}_b), \\ \frac{\hat{H}_{sB}}{\hbar} &= \hat{y}_b \otimes \hat{B},\end{aligned} \tag{H11}$$

with $\epsilon_d = C_pQ_b^{\text{zpf}}V_d/(2\hbar\bar{C})$ and $\hat{B} = C_pQ_b^{\text{zpf}}\hat{V}_B/(\hbar\bar{C})$.

Equation (H11) has the same structure as the single-ATS Hamiltonian of Ref. [7], with three differences: (i) the nonlinearity is the target form (H1), even under $\hat{x}_a \to -\hat{x}_a$ at all times; (ii) the memory does not appear in either the drive or the bath coupling ($u = 0$ exactly), i.e. it is a dark mode of the port; (iii) there is no linear drive on $\hat{x}_a$ for any η_p, so the memory counterpart of $g_{1,1}$ vanishes identically. The linear pump terms act on the buffer only and cancel at the first harmonic for the choice (H9). Two parity-protected operating points are therefore available: $\eta_p = 0$, the counterpart of the baseline single-ATS operation, and η_p given by Eq. (H9), the counterpart of its mitigated working point, which in addition has much smaller Stark and Kerr shifts.

Appendix I: Transient chaos

We integrate the classical mean-field equations of the memory and buffer amplitudes $a(t)$ and $b(t)$. The initial conditions form a uniform 10×10 grid in the memory phase space.

Each trajectory is sampled once per memory period $T_a = 2\pi/\omega_a$, this is the Poincaré map of the memory at period T_a. A periodic orbit of period T_a maps to a fixed point, a period-doubled orbit to two points, and a chaotic orbit to an extended set.

Each sampled trajectory is cut into $K = \lfloor N_s/M \rfloor$ consecutive windows of M points,

$$\mathcal{C}_k^{(j)} = \left\{a_n^{(j)} \,:\, (k-1)M \le n < kM\right\}, \qquad k = 1, \ldots, K. \tag{I1}$$

We will compute the entropy of these trajectory cuts and we start by cutting phase space in cells $\mathcal{B}_{lm}$. The same partition is used for every window, so that entropies

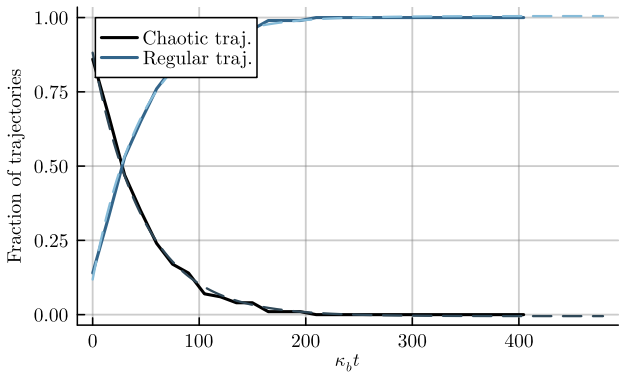


FIG. 18: Decay of the proportion of chaotic trajectories of the system. The solid lines are exponential fits.

can be compared. For each trajectory and window we compute the occupation probabilities and the Shannon entropy

$$p_{lm}^{(j,k)} = \frac{1}{M}\,\#\{a \in \mathcal{C}_k^{(j)} \,:\, a \in \mathcal{B}_{lm}\},$$
$$H_{j,k} = -\sum_{l,m} p_{lm}^{(j,k)} \ln p_{lm}^{(j,k)}, \tag{I2}$$

with $0 \ln 0 = 0$. By construction $0 \leq H_{j,k} \leq \ln M \simeq 5.30$.

- A window whose M points all fall in the same cell, such as a period-T_a orbit, gives $H = 0$.
- A period-$2T_a$ orbit gives $H = \ln 2$.
- A window whose points spread over M distinct cells, as for a chaotic trajectory, saturates at $H \simeq \ln M$.

Classification and fractions. A trajectory is counted as chaotic in window k if $H_{j,k} \geq H_\mathrm{c} > \ln 2$, and as regular if $H_{j,k} \leq H_\mathrm{c}$. The regular criterion selects attractors visited by at most two cells, i.e. period-T_a and period-$2T_a$ orbits. The plotted quantities are the fractions of the ensemble in each class,

$$f_\mathrm{c}(t_k) = \frac{1}{N}\,\#\{j : H_{j,k} \geq H_\mathrm{c}\}, \qquad f_\mathrm{r}(t_k) = \frac{1}{N}\,\#\{j : H_{j,k} \leq H_\mathrm{r}\}, \tag{I3}$$

shown as solid lines against $\kappa_b t_k$.

[TMD: Transient chaos literature, reference for exponential decay, Difficulty for the computation of Lyapunov exponents, work already on quantum transient chaos? Maybe the chaotic chaos is never transient?]

[1] C. Berke, E. Varvelis, S. Trebst, A. Altland, and D. P. DiVincenzo, Transmon platform for quantum computing challenged by chaotic fluctuations, Nat. Commun. **13**, 2495 (2022).

[2] J. Cohen, A. Petrescu, R. Shillito, and A. Blais, Reminiscence of Classical Chaos in Driven Transmons, PRX Quantum **4**, 020312 (2023).

[3] J. Chávez-Carlos, M. A. P. Reynoso, R. G. Cortiñas, I. García-Mata, V. S. Batista, F. Pérez-Bernal, D. A. Wisniacki, and L. F. Santos, Driving superconducting qubits into chaos, Quantum Sci. Technol. **10**, 015039 (2024).

[4] M. Mirrahimi, Z. Leghtas, V. V. Albert, S. Touzard, R. J. Schoelkopf, L. Jiang, and M. H. Devoret, Dynamically protected cat-qubits: a new paradigm for universal quantum computation, New J. Phys. **16**, 045014 (2014).

[5] R. Lescanne, M. Villiers, T. Peronnin, A. Sarlette, M. Delbecq, B. Huard, T. Kontos, M. Mirrahimi, and Z. Leghtas, Exponential suppression of bit-flips in a qubit encoded in an oscillator, arXiv 10.1038/s41567-020-0824-x (2019), 1907.11729.

[6] A. Marquet, A. Essig, J. Cohen, N. Cottet, A. Murani, E. Albertinale, S. Dupouy, A. Bienfait, T. Peronnin, S. Jezouin, R. Lescanne, and B. Huard, Autoparametric Resonance Extending the Bit-Flip Time of a Cat Qubit up to 0.3 s, Phys. Rev. X **14**, 021019 (2024).

[7] L. Carde, P. Rouchon, J. Cohen, and A. Petrescu, Flux-pump-induced degradation of T_1 for dissipative cat qubits, Phys. Rev. Appl. **23**, 024073 (2025).

[8] K. S. Dubovitskii, Bit-flip errors in dissipative cat qubits: Second-order perturbation theory, Phys. Rev. A **111**, 012617 (2025).

[9] T. Detroux, L. Renson, L. Masset, and G. Kerschen, The harmonic balance method for bifurcation analysis of large-scale nonlinear mechanical systems, Comput. Methods Appl. Mech. Eng. **296**, 18 (2015).

[10] Note that the Floquet multipliers consists either of two pairs of complex conjugate numbers or of 4 real coefficients, so that the dynamics is effectively described by n real numbers only in n-d.

[11] The blue islands are due to highly excited stable limit-cycle: due to chaos, the trajectory starting in the origin is quickly sent to these stable cycles and it appears as regular.

[12] S.-D. Börner, C. Berke, D. P. DiVincenzo, S. Trebst, and A. Altland, Classical chaos in quantum computers, Phys. Rev. Res. **6**, 033128 (2024).

[13] S. Pg, R. Modak, and S. Aravinda, Witnessing quantum chaos using observational entropy, arXiv 10.1103/PhysRevE.107.064204 (2022), 2212.01585.

[14] S. Pappalardi, A. Polkovnikov, and A. Silva, Quantum echo dynamics in the Sherrington-Kirkpatrick model, SciPost Phys. **9**, 021 (2020).
[15] L. Sá, P. Ribeiro, and T. Prosen, Complex Spacing Ratios: A Signature of Dissipative Quantum Chaos, Phys. Rev. X **10**, 021019 (2020).
[16] A. I. Larkin and Yu. N. Ovchinnikov, Quasiclassical Method in the Theory of Superconductivity, Soviet Journal of Experimental and Theoretical Physics **28**, 1200 (1969).
[17] I. García-Mata, R. A. Jalabert, and D. A. Wisniacki, Out-of-time-order correlators and quantum chaos, arXiv 10.4249/scholarpedia.55237 (2022), 2209.07965.
[18] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2006).
[19] The curve is not periodic because the system is chaotic.
[20] R. J. Bell and P. Dean, Atomic vibrations in vitreous silica, Discussions of the Faraday Society **50**, 55 (1970).
[21] F. Wegner, Inverse participation ratio in $2 + \varepsilon$ dimensions, Zeitschrift für Physik B Condensed Matter **36**, 209 (1980).
[22] B. Kramer and A. MacKinnon, Localization: theory and experiment, Reports on Progress in Physics **56**, 1469 (1993).
[23] F. M. Izrailev, Simple models of quantum chaos: Spectrum and eigenfunctions, Physics Reports **196**, 299 (1990).
[24] F. Haake, *Quantum Signatures of Chaos*, 3rd ed. (Springer, Berlin, Heidelberg, 2010).
[25] L. F. Santos and M. Rigol, Onset of quantum chaos in one-dimensional bosonic and fermionic systems and its relation to thermalization, Physical Review E **81**, 036206 (2010).
[26] F. Borgonovi, F. M. Izrailev, L. F. Santos, and V. G. Zelevinsky, Quantum chaos and thermalization in isolated systems of interacting particles, Physics Reports **626**, 1 (2016).
[27] X. You, J. A. Sauls, and J. Koch, Circuit quantization in the presence of time-dependent external flux, Phys. Rev. B **99**, 174512 (2019).